








Time Series Analysis

Seminar 4: Seasonality and forecasting: SARIMA, TBATS, Prophet



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **DHR**: Dynamic Harmonic Regression
- **DM**: Diebold–Mariano (test)
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- **MA**: Moving Average
- **MAE**: Mean Absolute Error
- **MAPE**: Mean Absolute Percentage Error
- **MASE**: Mean Absolute Scaled Error
- **MSE**: Mean Squared Error
- **PACF**: Partial AutoCorrelation Function
- **RMSE**: Root Mean Squared Error
- **SARIMA**: Seasonal ARIMA (model)
- **SCA**: Seasonally and Calendar Adjusted
- **SE**: Standard Error
- **TBATS**: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- **TSA**: Time Series Analysis
- **WD**: Working Days (regressor)
- **WN**: White Noise

Today's Question and Seminar Structure

- The question of the seminar
  - **how do we model, forecast and compare forecasts of series that repeat a pattern every year, week or day?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	seasonal differences and seasonal naive forecasts; SARIMA polynomials	Solved, Proposed	A1
A3, A4	the ACF of the airline model; a seasonal AR and moment estimates	Solved, Proposed	A3
A5, A6	MASE and the Diebold–Mariano test; combining two forecasts	Solved, Proposed	A5
B1, B2	SARIMA: Romanian GDP; industrial production with working days	Solved, Proposed	B1
B3, B4	cross-validation: electricity load; monthly inflation	Solved, Proposed	B3
C1, C2	tourism after the pandemic; what is wrong in an AI answer?	Proposed	B1, A5

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
Real GDP, not adjusted	Eurostat (namq_10_gdp)	quarterly	2000–2026
Industrial production	Eurostat (sts_inpr_m)	monthly, 2021 = 100	2010–2026
Electricity load	ENTSO-E	daily mean of hourly data	2022–2026
HICP, Romania	Eurostat (prc_hicp_minr)	monthly, 2015 = 100	2010–2026
Tourism nights	Eurostat (tour_occ_nim)	monthly	2005–2025

- GDP, production, prices and tourism are used as $100 \ln Y_t$ (Prerequisites 1/21)
- ENTSO-E: European Network of Transmission System Operators for Electricity; load = electricity consumed, in MW (hourly average)
- In the notebook: `fit_sarima`, `b1_gdp()`, `b3_load_cv()`; no account or key is needed

Prerequisites for Today (1/21): seasonality and the lag operator

- **Seasonality**: a pattern that repeats with a fixed **period** s
  - $s = 4$ (quarters), $s = 12$ (months), $s = 7$ (days of the week), $s = 24$ (hours of the day)
  - causes: weather, the calendar (working days, holidays), institutions (tax dates, school year)
- **Lag operator** L: shifts a series back in time

$$Ly_t = y_{t-1}, \qquad L^k y_t = y_{t-k}$$

  - k: the number of periods of the shift; polynomials in L are multiplied like ordinary polynomials
  - example: $(1 - L)(1 - L^4) = 1 - L - L^4 + L^5$
- **Logs times 100**: differences become growth rates in %

$$y_t = 100 \ln Y_t, \qquad y_t - y_{t-k} = 100 \ln \frac{Y_t}{Y_{t-k}} \approx 100 \frac{Y_t - Y_{t-k}}{Y_{t-k}}$$

  - Y_t: the original series (GDP, production, a price index); the approximation is good for changes below about 10%

Prerequisites for Today (2/21): seasonal differences

- **Seasonal difference**: each value minus the same season one year earlier

$$\Delta_s y_t = (1 - L^s)\, y_t = y_t - y_{t-s}$$

  - removes a seasonal pattern that repeats exactly; with $y_t = 100 \ln Y_t$ and $s = 4$: the year-on-year growth in %
- **Regular and seasonal difference** together

$$\Delta\Delta_s y_t = (1 - L)(1 - L^s)\, y_t = y_t - y_{t-1} - y_{t-s} + y_{t-s-1}$$

  - $\Delta = 1 - L$ also removes a trend; $\Delta\Delta_s y_t$: the change of the year-on-year growth
- **Seasonal naive forecast**: copies the same season of the last observed year

$$\hat{y}_{T+h|T} = y_{T+h-s}, \qquad h = 1, \ldots, s$$

  - T: the last observation; $\hat{y}_{T+h|T}$: the forecast h periods ahead, made at T
  - the benchmark every seasonal model must beat (Chapter 0)

Prerequisites for Today (3/21): a simulated quarterly series and its differences



- Simulated: $y_t = 100 + 0.8\,t +$ a seasonal pattern $(-6, 2, 5, -1)$ + noise, 40 quarters
- $\Delta_4 y_t$ fluctuates around $4 \times 0.8 = 3.2$: the season is gone, the trend becomes a constant
  - $\Delta\Delta_4 y_t$ fluctuates around 0: neither trend nor season; red: the seasonal naive forecast

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Prerequisites for Today (4/21): the ACF of an MA process

- **MA(q)** process: a weighted sum of current and past shocks

$$w_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \qquad \varepsilon_t \sim \text{WN}(0, \sigma^2)$$

  - WN: white noise, uncorrelated shocks with mean 0 and variance σ^2; θ_j: the weight of the shock of j periods ago

- **Autocovariance** $\gamma(k) = \text{Cov}(w_t, w_{t-k})$ and **autocorrelation** ρ_k

$$\gamma(k) = \sigma^2 \sum_{j=0}^{q-k} \theta_j \theta_{j+k} \quad (\theta_0 = 1), \qquad \rho_k = \frac{\gamma(k)}{\gamma(0)}$$

  - $\gamma(0) = \sigma^2(1 + \theta_1^2 + \cdots + \theta_q^2)$ is the variance; $\rho_k = 0$ for $k > q$: the ACF cuts off after lag q
  - MA(1): $\rho_1 = \theta/(1+\theta^2)$, always between -0.5 and 0.5

- **Sample ACF** $\hat{\rho}_k$: for white noise, approximately $N(0, 1/T)$
  - 95% band $\pm 1.96/\sqrt{T}$, T: the sample size; a bar outside the band is significant at 5%

Prerequisites for Today (5/21): PACF, invertibility and moment estimates

- **PACF** ϕ_{kk}: the last coefficient of an AR(k) regression

$$y_t = \phi_{k1} y_{t-1} + \cdots + \phi_{kk} y_{t-k} + e_t$$

  - the correlation of y_t and y_{t-k} after removing the lags in between; for an AR(p) it cuts off after lag p
- **Invertibility** of an MA(1): $|\theta| < 1$
  - the shocks ε_t can then be recovered from the past values of the series
  - θ and $1/\theta$ give the same ρ_1: we keep the root with $|\theta| < 1$
- **Method of moments**: set the theoretical ρ_1 equal to the sample value

$$\frac{\theta}{1+\theta^2} = \hat{\rho}_1 \iff \hat{\rho}_1\theta^2 - \theta + \hat{\rho}_1 = 0, \qquad \hat{\theta} = \frac{1 - \sqrt{1 - 4\hat{\rho}_1^2}}{2\hat{\rho}_1}$$

  - the two roots have product 1; the formula gives the invertible one; a solution exists only if $|\hat{\rho}_1| \leq 0.5$

Prerequisites for Today (6/21): SARIMA models

- **SARIMA**$(p, d, q)(P, D, Q)_s$: a regular and a seasonal ARMA, multiplied

$$\phi(L)\,\Phi(L^s)\,(1-L)^d(1-L^s)^D\,y_t = \theta(L)\,\Theta(L^s)\,\varepsilon_t$$

- Notation
  - $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$, $\theta(L) = 1 + \theta_1 L + \cdots + \theta_q L^q$: the regular AR and MA polynomials (Chapter 2)
  - $\Phi(L^s) = 1 - \Phi_1 L^s - \cdots - \Phi_P L^{Ps}$, $\Theta(L^s) = 1 + \Theta_1 L^s + \cdots + \Theta_Q L^{Qs}$: the seasonal AR and MA polynomials
  - d, D: the numbers of regular and of seasonal differences; ε_t: white noise with variance σ^2
- **Multiplied polynomials**: cross terms appear without new parameters
  - $(1 + \theta L)(1 + \Theta L^s) = 1 + \theta L + \Theta L^s + \theta\Theta L^{s+1}$
  - parameters: $p + q + P + Q$ coefficients and σ^2; a constant only when there is no difference

Prerequisites for Today (7/21): the airline model

- **Airline model** (Box and Jenkins): SARIMA$(0,1,1)(0,1,1)_s$

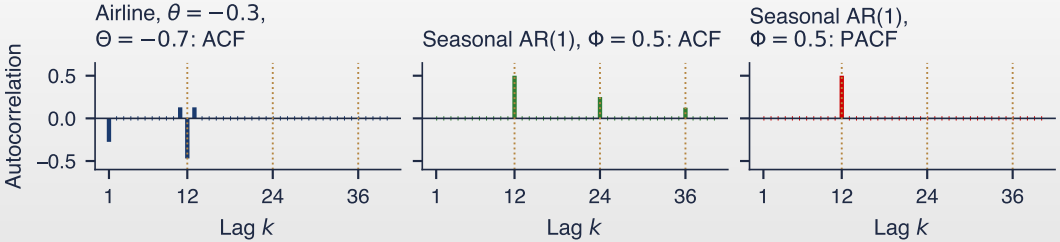
$$w_t = \Delta\Delta_s y_t = (1+\theta L)(1+\Theta L^s)\,\varepsilon_t = \varepsilon_t + \theta\varepsilon_{t-1} + \Theta\varepsilon_{t-s} + \theta\Theta\varepsilon_{t-s-1}$$

  - two parameters, θ and Θ; w_t is an MA$(s+1)$ with only four nonzero coefficients
- **ACF of** w_t, from the MA formula

$$\rho_1 = \frac{\theta}{1+\theta^2}, \qquad \rho_s = \frac{\Theta}{1+\Theta^2}, \qquad \rho_{s-1} = \rho_{s+1} = \rho_1\rho_s$$

  - all other autocorrelations are 0
- **The role of** Θ: how fast the seasonal pattern changes
  - the seasonal forecast is an exponentially weighted average of past patterns, with weight $1+\Theta$ on the newest year
  - $\Theta \to -1$: a fixed pattern; $\Theta = 0$: a pattern that changes freely from year to year

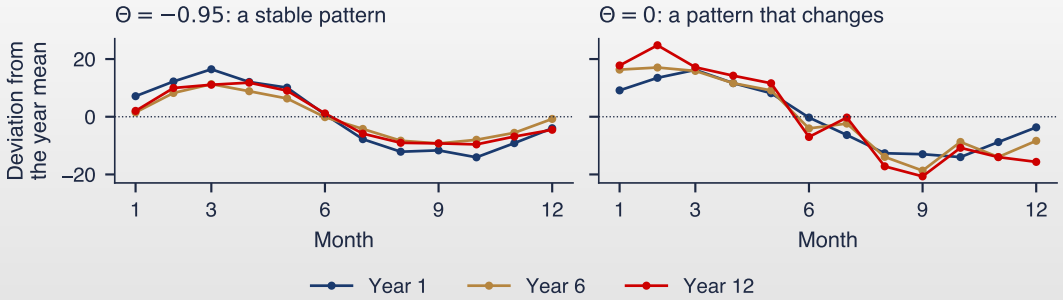
Prerequisites for Today (8/21): ACF and PACF of seasonal models



- Left: airline, $s = 12$: $\rho_{\mathbf{1}} = -0.28$, $\rho_{\mathbf{12}} = -0.47$ and the "satellites" $\rho_{\mathbf{11}} = \rho_{\mathbf{13}} = \rho_{\mathbf{1}}\rho_{\mathbf{12}} = 0.13$
- Middle and right: seasonal AR(1), $y_t = \Phi y_{t-\mathbf{12}} + \varepsilon_t$: $\rho_{\mathbf{12}k} = \Phi^k$, PACF a single spike at lag 12
  - ACF that cuts off after s and PACF that decays at $s, 2s, \ldots$: seasonal MA(1); the reverse: seasonal AR(1)

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Prerequisites for Today (9/21): a stable or a changing seasonal pattern



- Two simulated airline series, $\theta = -0.4$, the same shocks; each line: the months of one year minus the mean of that year
- $\Theta = -0.95$: years 1, 6 and 12 almost coincide (correlation 0.98 between year 1 and year 12)
  - $\Theta = 0$: the pattern drifts (correlation 0.93); the forecast then follows mostly the latest year

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Prerequisites for Today (10/21): estimation and diagnostics

- **Maximum likelihood**: $\hat{\theta}$, $\hat{\Theta}$ maximise the log-likelihood ℓ of the observed series
  - SE: the standard error of an estimate; $|\hat{\theta}/\mathrm{SE}| > 1.96$: significant at 5%
- **Ljung–Box** test on the residuals $\hat{\varepsilon}_t$

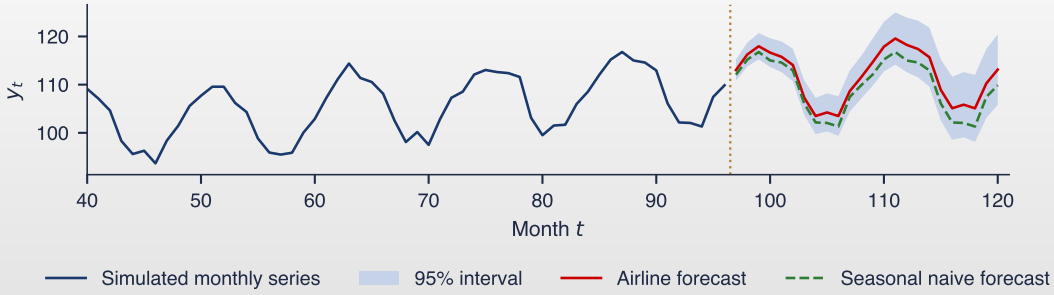
$$Q^*(m) = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k} \sim \chi^2(m - n_c)$$

  - $\hat{\rho}_k$: the residual ACF; m: the number of lags tested; $n_c = p + q + P + Q$: the ARMA coefficients (airline: $m - 2$)
  - H_0: the residuals are white noise; a small p-value: the model misses part of the dynamics
- **AICc**: the corrected Akaike information criterion

$$\mathrm{AICc} = -2\ell(\hat{\boldsymbol{\vartheta}}) + 2k + \frac{2k(k+1)}{T-k-1}$$

  - $\hat{\boldsymbol{\vartheta}}$: the estimated parameters; k: their number, σ^2 included; the smallest AICc wins (same data, same differences)

Prerequisites for Today (11/21): forecasts and their intervals



- 95% interval: $\hat{y}_{T+h|T} \pm 1.96\,\hat{\sigma}_h$; $\hat{\sigma}_h$: the standard deviation of the h-step forecast error
  - simulated airline series ($\theta = -0.4$, $\Theta = -0.6$): width 4.4 at $h = 1$, 8.8 at $h = 12$, 14.7 at $h = 24$
- Implied year-on-year growth of a forecast: $\hat{y}_{T+h|T} - y_{T+h-s}$ (in % when $y_t = 100 \ln Y_t$)

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Prerequisites for Today (12/21): regression with SARIMA errors

- Regressors known in advance, plus SARIMA dynamics in the error

$$y_t = \boldsymbol{\beta}^\top \mathbf{x}_t + u_t, \qquad u_t \sim \text{SARIMA}$$

  - $\mathbf{x}_t$: the vector of regressors; $\boldsymbol{\beta}$: their coefficients; $^\top$: the transpose
  - **working days**: the Monday–Friday days of the month that are not public holidays
  - **dummy**: 1 in a given period (a holiday, an outlier such as April 2020), 0 otherwise
- **Effect in %**: with $y_t = 100 \ln Y_t$, a coefficient β changes Y_t by

$$100\,(e^{\beta/100} - 1)\% \approx \beta\%$$

- **SCA** series: seasonally and calendar adjusted by the statistical office
  - compare an SCA quarter with the previous quarter; an unadjusted quarter only with the same quarter one year earlier

Prerequisites for Today (13/21): a test for monthly means

- **Month dummies**: $D_{mt} = 1$ if t falls in month m, 0 otherwise

$$\pi_t = \delta_1 D_{1t} + \cdots + \delta_{12} D_{12t} + e_t$$

  - $\pi_t = 100\,\Delta \ln P_t$: monthly inflation in %, P_t: the price index; δ_m: the mean of π_t in month m

- F **test** of H_0: $\delta_1 = \cdots = \delta_{12}$ (no seasonality in the mean)

$$F = \frac{(RSS_R - RSS_U)/q}{RSS_U/(T-k)} \sim F(q,\ T-k)$$

  - RSS_U: the residual sum of squares with the 12 dummies ($k = 12$ coefficients); RSS_R: with a constant only
  - $q = 11$ restrictions; a large F, with a small p-value: the month means differ

Prerequisites for Today (14/21): Fourier terms and dynamic harmonic regression

- **Fourier terms** for a period m

$$\sin\left(\frac{2\pi kt}{m}\right), \quad \cos\left(\frac{2\pi kt}{m}\right), \qquad k = 1, \ldots, K$$

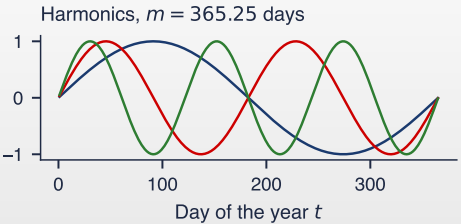
  - k: the harmonic, a wave with k cycles per period; K: the number of pairs
  - $2K$ coefficients instead of $m - 1$ dummies; m may be non-integer (365.25 days in a year)

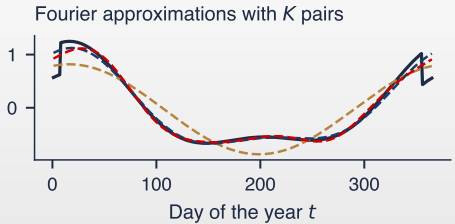
- **DHR** (dynamic harmonic regression): Fourier terms, calendar dummies and ARMA errors

$$y_t = \beta_0 + \sum_{k=1}^{K} \left[a_k \sin \frac{2\pi kt}{m} + b_k \cos \frac{2\pi kt}{m} \right] + \boldsymbol{\gamma}^\top \mathbf{x}_t + u_t$$

  - a_k, b_k: the weights of harmonic k; $\mathbf{x}_t$: weekday and holiday dummies, with coefficients $\boldsymbol{\gamma}$; u_t: ARMA errors
  - TBATS and Prophet (Lecture 4): automatic versions of the same idea, with several periods

Prerequisites for Today (15/21): harmonics and an annual pattern





$\sin(2\pi kt/m)$, $k = 1$ — $\sin(2\pi kt/m)$, $k = 3$ — $K = 1$ (2 coef.) — $K = 4$ (8 coef.)

$\sin(2\pi kt/m)$, $k = 2$ — Annual pattern — $K = 2$ (4 coef.)

- Left: the first three harmonics, with 1, 2 and 3 cycles per year
- Right: an annual pattern with a dip at the turn of the year, fitted by least squares
  - share of the variance explained: 86% with $K = 1$, 97% with $K = 2$, 98% with $K = 4$
  - the short holiday dip remains: it needs a dummy, not more harmonics

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Prerequisites for Today (16/21): measures of forecast accuracy

- Forecast errors $e_i = y_i - \hat{y}_i$, $i = 1, \ldots, n$, collected over all origins and horizons

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|e_i|, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}e_i^2}, \qquad \text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{e_i}{y_i}\right|$$

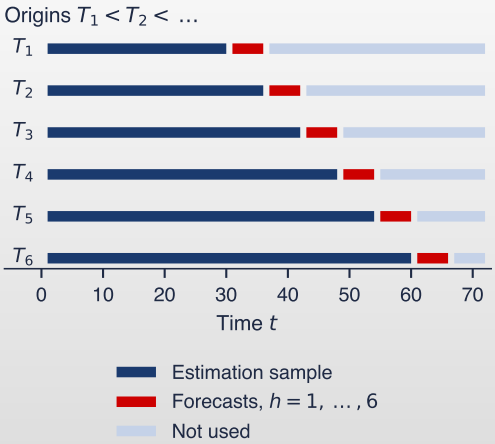
  - MAE (mean absolute error), RMSE (root mean squared error): in the units of y; RMSE penalises large errors more
  - MAPE (mean absolute percentage error): in %, meaningful only for y_i far from 0

- **MASE**: the MAE relative to the in-sample seasonal naive method

$$\text{MASE} = \frac{\text{MAE}}{\frac{1}{T-s}\sum_{t=s+1}^{T}|y_t - y_{t-s}|}$$

  - MASE (mean absolute scaled error) < 1: better than the in-sample seasonal naive forecast; comparable across series

Prerequisites for Today (17/21): time-series cross-validation



- **Rolling origin**: never use the future
  - at each origin T_i: estimate on the data up to T_i
  - then forecast $h = 1, \ldots, H$ steps ahead
- The errors of all origins and horizons
  - give the MAE, the RMSE and the MASE
  - here: 6 origins, 6 steps each, 36 errors
- Errors of the same origin overlap
  - they are correlated, so tests use one value per origin

Prerequisites for Today (18/21): the Diebold–Mariano test

- **Loss differential** of two forecasts of the same values

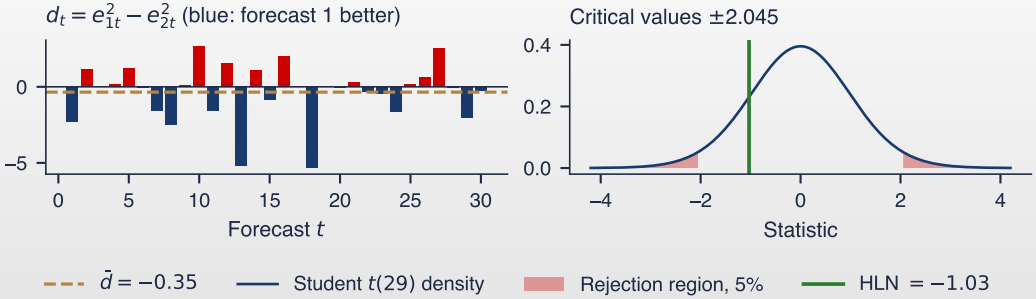
$$d_t = L(e_{1t}) - L(e_{2t}), \qquad L(e) = e^2 \text{ or } |e|$$

  - e_{1t}, e_{2t}: the errors of forecasts 1 and 2; $\bar{d}$: the mean of d_t over n values; H_0: $E[d_t] = 0$ (equal accuracy)
- **DM** statistic for one-step (or non-overlapping) losses, and the **HLN** correction

$$\mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2/n}}, \qquad \hat{\sigma}_d^2 = \frac{1}{n}\sum_t (d_t - \bar{d})^2, \qquad \mathrm{HLN} = \sqrt{\frac{n-1}{n}}\,\mathrm{DM}$$

  - HLN (Harvey, Leybourne and Newbold): a small-sample correction, compared with the Student $t(n-1)$ distribution
  - $\bar{d} < 0$: forecast 1 is more accurate; reject H_0 if $|\mathrm{HLN}| > t_{0.975}(n-1)$
  - 5% critical values: $t_{0.975}(5) = 2.571$, $t_{0.975}(7) = 2.365$

Prerequisites for Today (19/21): a Diebold–Mariano test on simulated errors



- Simulated errors of two forecasts, $n = 30$; mean squared errors 0.80 and 1.15
- $\bar{d} = -0.35$: forecast 1 is better on average, but d_t is very dispersed
  - HLN $= -1.03$ lies inside ± 2.045: equal accuracy is not rejected (p-value 0.31)

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Prerequisites for Today (20/21): forecast combination

- **Combined forecast**: a weighted average of two forecasts

$$f_c = w\, f_1 + (1 - w)\, f_2$$

  - f_1, f_2: the two forecasts; w: the weight of f_1
- **MSE** (mean squared error) of the combination, for unbiased forecasts

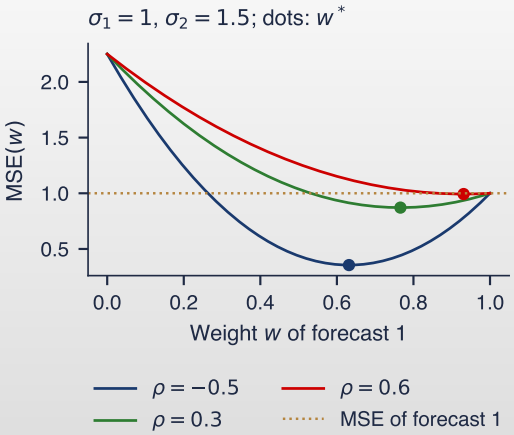
$$\text{MSE}(w) = w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$$

  - σ_1, σ_2: the standard deviations of the two errors; ρ: the correlation of the errors
- **Optimal weight**: the w that minimises MSE(w)

$$w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$$

  - the gain over the better forecast is large when ρ is small or negative

Prerequisites for Today (21/21): the gain from combining



- $\sigma_1 = 1$, $\sigma_2 = 1.5$: forecast 1 is better alone
  - $w = 1$: MSE $= \sigma_1^2 = 1$
- $\rho = -0.5$: $w^* = 0.63$, MSE 0.36
  - the errors cancel: a third of the MSE of forecast 1
- $\rho = 0.3$: $w^* = 0.77$, MSE 0.87
- $\rho = 0.6$: $w^* = 0.93$, MSE 0.99
  - similar errors: almost no gain

A1: seasonal differences by hand [Solved]

Task

- A quarterly series, $t = 1, \ldots, 9$: 10, 14, 16, 18, 12, 16, 19, 21, 14.
- 1. Compute $\Delta_4 y_t$ for $t = 5, \ldots, 9$.
- 2. Compute $\Delta\Delta_4 y_t$ for $t = 6, \ldots, 9$.
- 3. Expand $(1 - L)(1 - L^4)$ and check it on $t = 9$.
- 4. Give the seasonal naive forecasts for $t = 10, \ldots, 13$.
- Report: thirteen numbers and one sentence on what Δ_4 removes.

Solution

- 1. $\Delta_4 y$: $2, 2, 3, 3, 2$ (e.g. $12 - 10$, $16 - 14$)
- 2. $\Delta\Delta_4 y$: $0, 1, 0, -1$
- 3. $1 - L - L^4 + L^5$; $t = 9$: $14 - 21 - 12 + 18 = -1$
- 4. $\hat{y}_{10}, \ldots, \hat{y}_{13} = 16, 19, 21, 14$ (the last four values)
- Δ_4 removes a seasonal pattern that repeats exactly; the small, stable $\Delta_4 y$ shows the year-on-year growth.

Back Exercise list

A2: SARIMA polynomials [Proposed]

Solution discussed in class

Task

- Monthly data, $s = 12$. Model: A1.
- 1. Write the airline model SARIMA$(0,1,1)(0,1,1)_{\mathbf{12}}$ with $\theta = -0.4$, $\Theta = -0.6$ as an equation for y_t.
- 2. List the lags of y and of ε that appear, and the coefficient of $\varepsilon_{t-\mathbf{13}}$.
- 3. Expand $(1 - 0.5L)(1 - 0.3L^{\mathbf{12}})$ for SARIMA$(1,0,0)(1,0,0)_{\mathbf{12}}$.
- 4. Count the parameters of SARIMA$(1,1,1)(1,1,1)_{\mathbf{12}}$, including $\sigma^{\mathbf{2}}$.
- Report: two equations, two lists of lags and one count.

A3: the ACF of the airline model [Solved]

Task

- Quarterly airline model: $w_t = \Delta\Delta_{\mathbf{4}} y_t = (1 - 0.5L)(1 - 0.6L^{\mathbf{4}})\varepsilon_t$, $\varepsilon_t \sim \text{WN}(0, \sigma^{\mathbf{2}})$.
- 1. Write w_t as an MA(5) and list its nonzero coefficients.
- 2. Compute $\gamma(0)/\sigma^{\mathbf{2}}$.
- 3. Compute $\rho_{\mathbf{1}}, \ldots, \rho_{\mathbf{5}}$.
- 4. Say what the sample ACF of w_t should look like.
- Report: five autocorrelations and one sentence.

Solution

- 1. $w_t = \varepsilon_t - 0.5\varepsilon_{t-\mathbf{1}} - 0.6\varepsilon_{t-\mathbf{4}} + 0.3\varepsilon_{t-\mathbf{5}}$
- 2. $\gamma(0)/\sigma^{\mathbf{2}} = 1 + 0.25 + 0.36 + 0.09 = (1 + 0.25)(1 + 0.36) = 1.70$
- 3. $\rho_{\mathbf{1}} = -0.5/1.25 = -0.400$; $\rho_{\mathbf{4}} = -0.6/1.36 = -0.441$; $\rho_{\mathbf{3}} = \rho_{\mathbf{5}} = \rho_{\mathbf{1}}\rho_{\mathbf{4}} = 0.176$; $\rho_{\mathbf{2}} = 0$
- 4. Spikes at lags 1 and 4 with small "satellites" at 3 and 5; nothing after lag 5: the signature of the airline model.

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A4: a seasonal AR and moment estimates [Proposed]

Solution discussed in class

Task

- Model: A3.
- 1. For $y_t = 0.7y_{t-4} + \varepsilon_t$, compute ρ_4, ρ_8, ρ_{12} and say which other lags are zero.
- 2. Describe its PACF.
- 3. The sample ACF of $\Delta\Delta_{12}y_t$ is -0.40 at lag 1, -0.45 at lag 12, small at lags 11 and 13, and inside the band elsewhere; propose a model.
- 4. Estimate θ and Θ from ρ_1 and ρ_{12}, keeping the invertible roots.
- Report: three autocorrelations, a model and two estimates.

A5: MASE and the Diebold–Mariano test [Solved]

Task

- Six non-overlapping forecast windows; errors of method 1: $0.4, -0.6, 0.9, -0.3, 0.5, -0.8$; of method 2: $1.1, -0.9, 0.7, -1.2, 0.8, -1.0$; the in-sample MAE of the seasonal naive method is 1.
- 1. Compute the MAE and the MASE of both methods.
- 2. Compute $d_t = e_{1t}^2 - e_{2t}^2$, $\bar{d}$ and $\hat{\sigma}_{\bar{d}}^2$.
- 3. Compute DM and HLN.
- 4. Decide at 5% with $t(5)$.
- Report: four accuracy measures, the test statistic and a decision.

Solution

- 1. MAE: 0.583 and 0.950; MASE: the same (scale 1): both below 1, method 1 better
- 2. d: $-1.05, -0.45, 0.32, -1.35, -0.39, -0.36$; $\bar{d} = -0.547$; $\hat{\sigma}_{\bar{d}}^2 = 0.2864$
- 3. $\mathrm{DM} = -0.547/\sqrt{0.2864/6} = -2.50$; $\mathrm{HLN} = 0.913 \cdot (-2.50) = -2.28$
- 4. $|-2.28| < 2.571$ ($\mathrm{p} = 0.071$): equal accuracy is not rejected; six windows are too few to prove the visible gain.

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A6: two forecasts and their combination [Proposed]

Solution discussed in class

Task

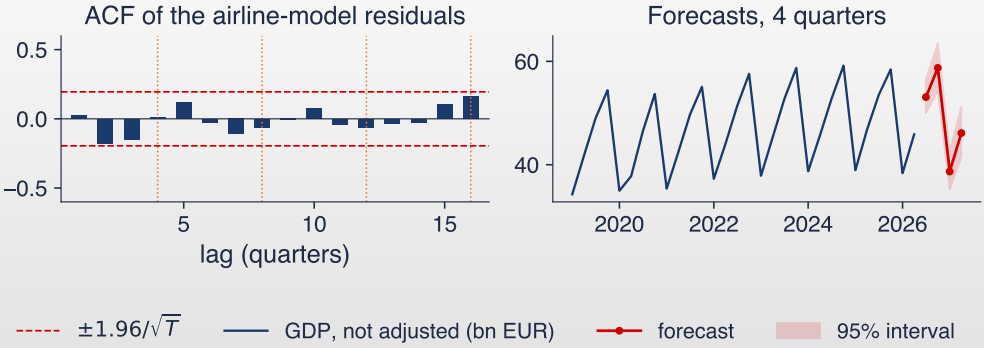
- Eight windows; errors e_1: 2.0, −1.5, 1.0, −2.5, 3.0, 0.5, −1.0, 1.5; e_2: 1.0, −2.5, 2.0, −0.5, 1.5, 2.0, −2.0, 0.5. Model: A5.
- 1. Compute the MAE of both methods and the DM and HLN statistics with squared losses.
- 2. Decide at 5% with $t(7)$.
- 3. The equal-weight combination has errors $(e_1 + e_2)/2$: compute its MAE and RMSE.
- 4. For unbiased forecasts with $\sigma_1 = 2$, $\sigma_2 = 3$, $\rho = 0.3$, compute MSE(0.5), w^* and MSE(w^*).
- Report: six numbers, a decision and three numbers for the combination.

B1: the airline model for Romanian GDP [Solved]

- **Question**: how stable is the seasonal pattern of Romanian GDP, and what does the airline model forecast for the next four quarters?
- **Inputs**: Eurostat, real GDP not adjusted, 2000Q1–2026Q2, $T = 106$; $y_t = 100 \ln Y_t$
- **Tasks**
  1. Estimate SARIMA$(0,1,1)(0,1,1)_{\mathbf{4}}$ by maximum likelihood and report $\hat{\theta}$, $\hat{\Theta}$ and their standard errors.
  2. Test the residuals with Ljung–Box $Q^*(8)$ on $8 - 2 = 6$ degrees of freedom.
  3. Forecast the next four quarters, in bn EUR, with 95% intervals.
  4. Compute the implied year-on-year growth for each forecast quarter.
  5. Interpretation: what does $\hat{\Theta}$ say about how fast the seasonal pattern of GDP changes?
- **Report**: two estimates with standard errors, a test, four forecasts with intervals and one sentence
- Notebook: section B1

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B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

- $\hat{\theta} = -0.171$ (0.109): not significant; $\hat{\Theta} = -0.585$ (0.072); $\hat{\sigma} = 3.05\%$; $Q^*(8) = 9.30$, p = 0.157: no residual autocorrelation
- Forecasts from 2026Q3: 53.1, 58.7, 38.7, 46.1 bn EUR; first interval $[50.0, 56.4]$, fourth $[41.5, 51.2]$
- Implied year-on-year growth: -0.84%, 0.52%, 0.83%, 0.53%
- Interpretation: the seasonal forecast is an exponentially weighted average of past seasonal patterns, with weight $1 + \hat{\Theta} = 0.42$ on the newest year; $\Theta = -1$ would mean a fixed pattern, $\Theta = 0$ a pattern that changes freely: the pattern of GDP evolves slowly

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B2: industrial production and working days [Proposed]

- **Question**: does the number of working days explain part of the monthly movements of Romanian industrial production?
- **Inputs**: Eurostat, industrial production not adjusted, 2010–July 2026, $y_t = 100 \ln Y_t$; dummies for April and May 2020; model: B1
- **Tasks**
  1. Estimate the airline model with and without the number of working days as a regressor, and compare AICc and $Q^*(24)$.
  2. Add two alternatives with working days, $(2,1,0)(0,1,1)_{\mathbf{12}}$ and $(1,1,1)(0,1,1)_{\mathbf{12}}$, and choose a model.
  3. Report the working-day coefficient with its standard error, and forecast the next 12 months.
  4. Interpretation: why must the working-day effect be removed before two months are compared?
- **Report**: a table of AICc and p-values, one coefficient, the forecast chart and one sentence
- Notebook: section B2

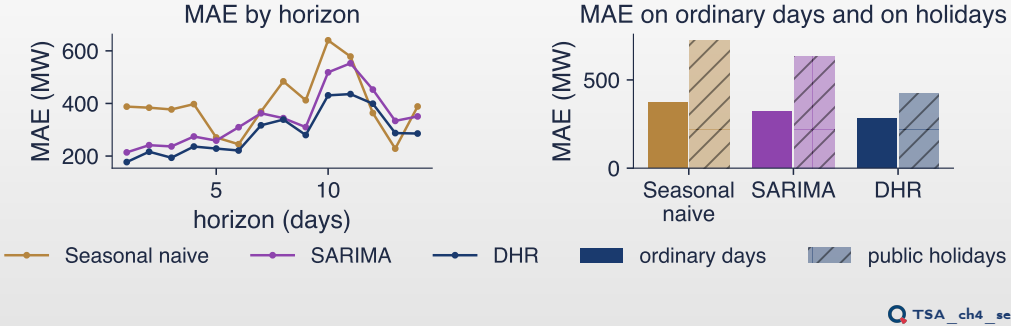
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B3: cross-validation for daily electricity load [Solved]

- **Question**: does a regression with Fourier terms and holiday dummies forecast daily load better than SARIMA and the seasonal naive method?
- **Inputs**: ENTSO-E, daily mean load in Romania (GW), 2022–2026; 13 origins every 28 days from 30 June 2025 to 1 June 2026, horizon 14 days
- **Tasks**
  1. At each origin, fit the seasonal naive method (period 7), SARIMA$(1,0,1)(0,1,1)_7$ and a DHR (4 annual Fourier pairs, weekday and holiday dummies, ARMA(1,1) errors).
  2. Compute the MAE and the MASE of each method over all origins and horizons.
  3. Test DHR against SARIMA with the DM test on the window-average squared errors (one value per origin), with the HLN correction.
  4. Compare the MAE on public holidays with the MAE on ordinary days.
  5. Interpretation: why is the gain of the DHR largest on public holidays?
- **Report**: a table of MAE and MASE, one test, the chart and one sentence
- Notebook: section B3

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B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

Method	MAE (MW)	MASE	holidays	other days
Seasonal naive	395	1.28	723	372
SARIMA	340	1.10	630	319
DHR	**289**	**0.94**	422	280

- MASE scale: in-sample MAE of the weekly seasonal naive method, 309 MW; DM (HLN), DHR against SARIMA: -1.96, $p = 0.073$; against the seasonal naive method: -2.73, $p = 0.018$
- Interpretation: SARIMA and the seasonal naive method only know "one week ago"; a holiday that falls on a weekday looks to them like a normal working day, while the DHR has dummies that remove 0.37 GW on a public holiday and 0.56 GW on the Easter days

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B4: seasonality in monthly inflation [Proposed]

- **Question**: is the seasonal pattern of Romanian monthly inflation strong enough to improve one-month forecasts?
- **Inputs**: HICP, Romania, 2010–2026; $\pi_t = 100\,\Delta \ln P_t$ (% per month); model: B3
- **Tasks**
  1. Regress π_t (2010–2019) on 12 month dummies and test equal month means with an F-test.
  2. Estimate the airline model for $100 \ln P_t$ and report $\hat{\theta}$ and $\hat{\Theta}$.
  3. From monthly origins since January 2016, forecast next month's π with SARIMA, the seasonal naive method and the mean of the last 12 months; compare the MAE.
  4. Test SARIMA against both benchmarks with the DM test (squared errors).
  5. Interpretation: is the seasonal pattern strong enough to improve one-month forecasts?
- **Report**: an F-test, two estimates, three MAE values, two DM tests and one sentence
- Notebook: section B4

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C1: tourism after the pandemic [Proposed]

- **Question**: which training sample gives better forecasts of tourism nights for 2023–2025: data up to 2019, or data up to 2022?
- **Inputs**: Eurostat, nights in tourist accommodation, Romania, monthly, 2005–2025; model: B1
- **Tasks**
  1. Fit the airline model to $\ln y_t$ up to December 2019 and forecast 2023–2025.
  2. Repeat with data up to December 2022.
  3. Compare both with the seasonal naive forecast that repeats 2022, using the MAPE (mean absolute percentage error).
  4. Propose a better treatment of 2020–2021 (intervention dummies, a shorter sample, TBATS or Prophet) and sketch it as a team project.
- **Report**: three MAPE values, the chart and a project plan
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant about seasonal models. The answer:
- (a) SARIMA(0,1,1)(0,1,1)4 has four MA parameters, at lags 1, 4 and 5 plus the noise variance.
- (b) If the ACF of the differenced monthly series has one spike at lag 12 and the PACF decays at 12, 24, 36, use a seasonal AR(1).
- (c) Unadjusted Romanian GDP fell by about 34% in the first quarter of 2026 compared with the fourth quarter of 2025: a deep recession.
- (d) DM = −2.3 with d = L(e1) - L(e2) shows that forecast 2 is significantly more accurate.
- (e) TBATS with periods 7 and 365.25 captures the Easter dip of electricity load.
- (f) The equal-weight average of two unbiased forecasts never has a larger MSE than the average of their two MSEs.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- Δ_s removes a stable seasonal pattern; $\Delta\Delta_s$ also removes the trend; the seasonal naive forecast is the benchmark
- SARIMA multiplies a regular and a seasonal polynomial: the airline model has two parameters and ACF spikes at 1, $s-1$, s, $s+1$
- Calendar effects (working days, Easter, public holidays) are known in advance: put them in as regressors
- Compare forecasts out of sample, with rolling origins, MASE and the DM test; a visible gain is not always a significant one
- Combining forecasts is cheap insurance: it is rarely the worst and often near the best

After the seminar

- Details: Chapter 4: the derivation of today's formulas; seasonal unit roots, seasonal adjustment (X-13ARIMA-SEATS), multiple seasonality, TBATS and Prophet
- TBATS and Prophet are optional installs in the notebook
- C1 can grow into a team project: forecasting tourism or electricity load in Romania with a rolling-origin comparison of several models
- Reading: Huang and Petukhina (2022), Ch. 4–5; FPP3, Section 9.9; FPP3, Section 5.10; Diebold and Mariano (1995)

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